



Huawei HCIA-AI

1. Module Information

Module Title	HCIA-AI
Duration	15 Sessions
Total Hours	45 Hours

2. Module Description

The HCIA-AI program is structured into six modules: AI Overview , which introduces AI concepts, history, applications, and foundation models; Machine Learning Overview, covering supervised, unsupervised, and reinforcement learning with core algorithms and evaluation methods; Basics of Deep Learning and Foundation Models, explaining the evolution of neural networks, deep learning fundamentals, and Transformer-based architectures with applications in NLP and computer vision; AI Development Framework, focused on Framework basics, model construction, training, and deployment; AI Business Process Overview , addressing the AI lifecycle, enterprise adoption, and large model integration in business processes; and Cutting-edge AI Applications , exploring modern AI use cases such as voice assistants, smart homes, intelligent vehicles, and robotics, with a simplified view of quantum computing and future AI trends.

3. Learning Outcomes

- 1 Understand and explain the fundamental concepts of Artificial Intelligence, Machine Learning, Deep Learning, and foundation models, along with their applications in real-world domains.
- 2 Apply core machine learning and deep learning algorithms using frameworks such as PyTorch/Tensorflow to build, train, and evaluate AI models.
- 3 Demonstrate practical skills in developing neural networks, implementing experiments, and deploying AI models in real-world scenarios.



- 4 Analyze and manage the AI project lifecycle, including business processes, data handling, model deployment, and enterprise integration of large models.
- 5 Evaluate emerging AI applications and trends, including intelligent assistants, smart systems, robotics, and other cutting-edge technologies, while recognizing future directions in AI.

4. Session Breakdown

Session	Topic	Hours	LOs
1	Introduction to AI – History, definitions, AI vs ML vs DL, key milestones.	3	1
2	AI Applications & Foundation Models – AI in daily life, industry use cases, basics of foundation models.	3	1
3,4	Supervised Learning – Regression (linear, logistic), decision trees, evaluation metrics.	6	2
5	Unsupervised Learning – Clustering (K-means, hierarchical), dimensionality reduction.	3	3
6,7	Neural Networks Basics – Perceptron, multilayer perceptrons, activation functions, Forward pass, backpropagation, optimizers.	6	4
8	Convolutional Neural Networks (CNNs) – Image recognition and applications.	3	4
9	Recurrent Neural Networks (RNNs) & LSTMs – Sequential data processing.	3	4
10	Transformers & Foundation Models – Architecture, self-attention, LLM basics.	3	3,4
11,12	Framework Basics – Tensors, autograd, modules, training workflow.	6	2,3,4



13	AI Business Lifecycle – Requirement analysis, data processing, modeling, deployment, monitoring; integration of large models into enterprises.	3	5
14	Modern AI Applications & Trends – Voice assistants, smart homes, intelligent vehicles, robotics, simplified view of quantum computing, and future AI directions.	3	5
15	Exam	3	1,2,3,4,5

5. Teaching Methods

#	Method
1	Guided hands-on labs .
2	Team-based deployment exercises.
3	Case studies of ML/DL applications.

6. Assessment Methods

#	Method	Marks
1	Quizzes and mooc Exam	40%
2	Hands-on Huawei Labs	30%
3	Project	30%